Machine Learning-Based Bias Correction of a Low-Cost MEMS Gyroscope for Dead-Reckoning Odometry in an Ackermann Mobile Robot

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***Abstract*—Autonomous navigation in low-cost mobile robots relies on MEMS-based inertial measurement units (IMUs), whose bias, noise, and thermal drift accumulate during integration and degrade position estimation. This work evaluates machine learning to correct the bias in the raw angular velocity of an Ackermann-steered mobile robot to improve odometry. Random Forest and LSTM models were trained to predict the residual between a Vicon motion-capture reference and the raw gyroscope reading, using gyroscope and magnetometer data as inputs. Both were compared against three classical correction methods and the uncorrected signal over three closed trajectories (circle, oval, and figure-eight). Both models reduced the instantaneous RMSE of the residual, but this did not consistently reduce the integrated position error. Random Forest and LSTM improved position accuracy on constant-curvature trajectories under pure dead-reckoning, without a model-based fusion filter such as an EKF, while classical methods degraded it even with EKF fusion. Both degraded sharply on the figure-eight trajectory, showing that reducing instantaneous error alone does not guarantee improved navigation accuracy.**

***Index Terms*—low-cost sensors, autonomous navigation, machine learning, gyroscope bias correction, dead reckoning, sensor fusion, LSTM, Random Forest**

I. Introduction

Autonomous navigation is a core capability of modern mobile robotics, allowing ground, aerial, and marine vehicles to move without human intervention by fusing information from multiple onboard sensors, such as inertial measurement units (IMUs), Global Navigation Satellite System (GNSS) receivers, cameras, and Light Detection and Ranging (LiDAR) sensors, to estimate position, orientation, velocity, and acceleration [1]. The quality of the resulting navigation estimate is directly tied to the quality of these measurements: IMUs provide high-rate acceleration and angular velocity data [2], while GNSS and LiDAR provide absolute position references and spatial maps of the environment, respectively.

High-grade sensors offer excellent stability and accuracy but are costly and power-demanding, which limits their use in low-cost robotic platforms. Low-cost MEMS-based IMUs, in particular, suffer from cumulative drift and stochastic noise that progressively degrade position estimates obtained through dead-reckoning [3]. Traditional model-based fusion approaches, such as the Extended Kalman Filter (EKF) or the Invariant Extended Kalman Filter (InEKF), can mitigate these errors, but their performance depends on the validity of linear or near-linear noise assumptions that are frequently violated by low-cost, temperature- and vibration-sensitive sensors [3], [4].

In recent years, artificial intelligence has emerged as a promising alternative to strengthen the correction of low-cost inertial sensors, learning nonlinear relationships between raw measurements and reference signals without an explicit noise model [5], [6]. For example, [4] combined deep learning with an InEKF to improve dead-reckoning odometry from a low-cost IMU, while [7] showed that super-resolution techniques can reconstruct LiDAR point clouds comparable to those of higher-end sensors. Still, gaps remain in the literature regarding models that improve low-cost sensor accuracy in an integral, position-aware manner, and regarding the availability of experimental datasets to train and validate such models under real operating conditions [8], [9].

This paper addresses the following research question: *how can a machine learning algorithm be designed to improve the accuracy of low-cost sensors used in autonomous navigation systems?* We focus specifically on the correction of the yaw-rate channel (ω_z) of a low-cost MEMS gyroscope mounted on an Ackermann-steered mobile robot, and we evaluate the effect of this correction not only on the instantaneous signal error, but critically on the resulting integrated position error, using an external motion-capture system as ground truth.

The main contributions of this work are: (i) an experimental comparison of Random Forest and causal LSTM models for learning-based residual correction of a low-cost MEMS gyroscope on a real Ackermann-steered mobile robot, using Vicon motion capture as ground truth; (ii) a leave-one-recording-out validation protocol that evaluates generalization across independent recordings, together with a comparison against three classical correction methods; and (iii) experimental evidence that reducing the instantaneous angular-rate residual does not necessarily translate into improved dead-reckoning position

accuracy, particularly on trajectories with frequent turning-direction changes.

II. Related Work

Model-based fusion schemes (KF, EKF, InEKF) remain the most widely used approach for inertial navigation, combining IMU, encoder, LiDAR, and/or GNSS measurements under an assumed statistical noise model [4]. When these assumptions are violated by nonlinear, temperature-dependent sensor behavior, learning-based and hybrid approaches have been proposed instead. Deep learning architectures, particularly convolutional neural networks (CNNs) for spatial data and recurrent networks such as LSTM for sequential IMU or encoder signals, have been used to learn hierarchical representations of sensor error [5]. Physics-informed hybrid models combine known equations of motion with neural networks that learn unmodeled residual components, improving interpretability while reducing training-data requirements [6]. Closer to the present work, [10] used recurrent networks jointly with estimation filters to adapt noise parameters and bound inertial drift in vehicle navigation, and [11] combined model-based and deep-learning components to process low-cost IMU data in wheeled mobile robot dead-reckoning. These precedents motivate the choice of Random Forest and LSTM as complementary regression paradigms in this work: one relying on explicit, hand-engineered temporal features, and the other learning temporal dependencies end-to-end directly from the raw sequence.

III. Methodology

A. Platform and Sensor Selection

Experiments were carried out on a ROSMASTER A1 mobile robot, an Ackermann-steered (car-like) platform equipped with an onboard NVIDIA Jetson Orin Nano, an IMU, wheel encoders, a monocular RGB camera, and a 2D LiDAR. Among the available sensing modalities (GNSS, LiDAR, RGB camera, wheel encoders, and IMU), the yaw-rate channel of the gyroscope, ω_z, was selected as the object of characterization and correction. GNSS was discarded due to the indoor test environment; LiDAR and the RGB camera were discarded because their use would have required implementing an independent odometry, localization, or SLAM pipeline, shifting the scope of the work toward spatial perception rather than sensor-level correction; wheel encoders were retained as an auxiliary input to the EKF baseline, but not selected as an independent object of AI-based correction, since odometric error also depends on mechanical parameters (wheel radius, wheelbase, slip) beyond the encoder signal itself.

For an Ackermann platform moving on a horizontal plane, the yaw rate is related to the vehicle's longitudinal velocity v, wheelbase L, and steering angle δ by

$$\dot{\psi} = \frac{v}{L}\tan(\delta), \tag{1}$$

under ideal rolling conditions and no lateral slip [12], [13]. In practice, a low-cost MEMS gyroscope measurement can be modeled as

$$\omega_{z,m} = \omega_z + b_z + n_z + e_z, \tag{2}$$

where b_z is the gyroscope bias, n_z is random noise, and e_z groups scale-factor variations, thermal effects, and other unmodeled components. Because orientation is obtained by integrating $\omega_{z,m}$ over time, even a small constant bias introduces a yaw error that grows approximately linearly with time, $e_\psi(t) \approx b_z t$ [14], motivating a data-driven correction of ω_z rather than relying solely on a static calibration constant.

B. Ground-Truth Reference and Target Definition

A Vicon motion-capture system, using infrared cameras and reflective markers mounted on the robot, was used as an external ground-truth reference, independent of any onboard sensor integration. IMU and Vicon streams were synchronized using bag-level timestamps with a common time origin $t_0 = \min(t_{\text{IMU}}[0], t_{\text{Vicon}}[0])$. The reference yaw angle ψ_{Vicon} was unwrapped at the full Vicon sampling rate ($\sim$100 Hz) prior to interpolation to the IMU timestamps ($\sim$10 Hz), and differentiated to obtain a reference angular velocity $\omega_{\text{Vicon}}(t)$, after removing spurious velocity spikes exceeding 2.0 rad/s or corresponding to near-duplicate timestamps ($dt_{\min} = 0.003$ s).

The learning target was defined as the residual between the reference and the raw gyroscope reading:

$$r(t) = \omega_{\text{Vicon}}(t) - \omega_{\text{gyro_z}}(t), \tag{3}$$

so that both models solve a regression problem over $r(t)$ rather than reconstructing the reference signal directly. The corrected signal is then obtained as

$$\omega^{\text{corr}}_{\text{gyro_z}}(t) = \omega_{\text{gyro_z}}(t) + \hat{r}(t). \tag{4}$$

C. Dataset

Data were collected on a modular, reconfigurable indoor track inside the Vicon coverage area, over three closed trajectories with distinct turning dynamics: a circle (constant curvature, single turning direction), an oval (alternating straight and curved segments), and a figure-eight (two lobes combining both turning directions within a single run). All trajectories were teleoperated at a constant velocity of 0.6 m/s, in both clockwise (CW) and counter-clockwise (CCW) directions, starting from a fixed floor marker to ensure consistent initial conditions.

After a data-quality diagnostic that discarded recordings with non-simultaneous or clustered communication gaps across topics, the training dataset consisted of **16 confirmed recordings** (6 circle, 6 oval, 4 figure-eight; 8 CW / 8 CCW), with an average duration of $\sim$292 s each, totaling 47,033 synchronized gyroscope–magnetometer samples at $\sim$10 Hz.

D. Model Inputs

For **Random Forest**, input features were engineered explicitly from the raw signal $\omega_{\text{gyro_z}}(t)$: temporal lags at 1, 2, 3, and 5 samples, rolling mean and standard deviation over 5, 15, and

30-sample windows, and the raw magnetometer components (mag_x, mag_y, mag_z), their magnitude, and its first-order lag. All windowed features respected recording segment boundaries to avoid leaking information across previously removed data gaps.

For LSTM, causal windows of 50 consecutive samples ($\sim$5 s at 10 Hz) were used instead of tabular features. Each sample in the window contained only the raw $\omega_{\text{gyro_z}}(t)$ value and the magnetometer magnitude $\sqrt{\text{mag}_x^2 + \text{mag}_y^2 + \text{mag}_z^2}$, letting the network learn temporal dependencies directly rather than relying on precomputed statistics.

The magnetometer magnitude was included in both models because it is an orientation-invariant, physically independent source of information: gyroscope bias in low-cost MEMS sensors is not purely deterministic but varies with temperature, vibration, and instantaneous operating conditions that are not observable from $\omega_{\text{gyro_z}}(t)$ alone [3], [15]. Although both models therefore combine gyroscope and magnetometer measurements at their input, the correction is learned as a data-driven regression on the gyroscope residual and does not involve a model-based fusion filter (e.g., EKF or complementary filter); in this paper, the term *fusion* refers exclusively to the latter.

E. Training and Validation Protocol

Both models were trained and evaluated using grouped, leave-one-bag-out cross-validation (`GroupKFold`), with the recording identifier as the grouping variable, so that no fold contains temporally correlated samples from the same recording in both training and validation. Out-of-fold predictions were consolidated across all 16 folds for evaluation. Observations with a reference residual exceeding 2.0 rad/s in magnitude ($\sim$0.1% of samples) were removed prior to Random Forest training as non-physical artifacts.

The hyperparameters of both models are summarized in Table I. They were not obtained through an exhaustive search; instead, they were fixed a priori according to explicit criteria: the moderate dataset size ($\sim$47k samples) and low input dimensionality; the need to limit overfitting to individual recordings (bounded tree depth, minimum leaf size, dropout, and early stopping); the 4 GB GPU memory available for training, which constrained the batch size; and strict causality, required for real-time deployment on the robot. These values were then assessed through the leave-one-bag-out protocol described above.

F. Estimation Pipeline and Evaluation Protocol

Robot orientation was obtained by direct integration of the corrected gyroscope signal, $\psi(t) = \psi(t-1) + \omega_{\text{gyro_z_corr}}(t) \cdot dt$, without an effective contribution from an onboard Madgwick AHRS filter (kept active only to expose an optional absolute-yaw reference, disabled in all recordings used here). For the four classical reference methods (raw IMU, constant-mean, 15-sample moving average, and 1D Kalman filter), this orientation estimate was additionally fused with wheel-encoder odometry through an EKF. The two machine learning methods (Random Forest and LSTM) were evaluated under pure dead-reckoning, *without* this additional fusion stage — a comparatively harder test, since no fusion mechanism can compensate for an imperfect gyroscope correction.

TABLE I
MODEL HYPERPARAMETERS

	Random Forest	**LSTM**
Structure	200 trees, max. depth 15, min. samples per leaf 5	2 layers, 64 hidden units, causal (unidirectional)
Input	Engineered features (Sec. III-D)	Window of 50 samples
Regularization	Depth and leaf limits	Dropout 0.2, early stopping (patience 5)
Optimization	–	Adam, lr 10^{-3}, MSE loss, batch 64, max. 30 epochs

Estimated trajectories were compared against the Vicon reference using Procrustes alignment (jointly optimizing translation, rotation, and scale), which proved more robust than a fixed initial-heading alignment. Position error at each instant was computed as the Euclidean distance between the aligned reference and the estimated trajectory, from which the position RMSE was obtained for each method and trajectory. Each classical method was evaluated over 9 recordings (3 trajectories $\times$ 3 repetitions); each AI method was evaluated using a dual odometry node publishing raw and corrected odometry simultaneously from the same physical run, over $N = 4$ repetitions per trajectory. A straight-line trajectory and an exploratory zigzag trajectory were additionally used to complement the closed-trajectory results.

IV. RESULTS

A. Instantaneous Correction of the Gyroscope Residual

Table II summarizes the leave-one-bag-out performance of both models on the instantaneous residual of $\omega_{\text{gyro_z}}$. Both models achieved comparable average RMSE improvements over the raw signal, with Random Forest slightly ahead in magnitude (12.7% vs. 11.2%) but LSTM more consistent, with no degraded recordings out of 16, versus one degraded recording for Random Forest. Both improvements were statistically significant (Wilcoxon paired test, $p = 0.00006$ for Random Forest and $p = 0.00003$ for LSTM).

TABLE II
INSTANTANEOUS PERFORMANCE ON THE GYRO_Z RESIDUAL (16-FOLD LEAVE-ONE-BAG-OUT)

Model	**RMSE (rad/s)**	**Avg. impr. (%)**	**Range (%)**	**Degr.**
Random Forest	0.294 ± 0.049	12.7	-0.9 to 22.7	1/16
LSTM	0.299 ± 0.047	11.2	1.7 to 19.6	0/16

B. Position Error of Classical Reference Methods

Table III shows that all three classical correction methods, when evaluated through the EKF fusion pipeline, produced a *higher* average position RMSE than the uncorrected raw IMU. The 15-sample moving average degraded position accuracy the most (-166.8%), consistent with the phase lag introduced by a fixed-window filter; the 1D Kalman filter was the only classical method that outperformed the constant-mean correction, though it still degraded position relative to the raw signal.

TABLE III
POSITION RMSE OF CLASSICAL REFERENCE METHODS VS. RAW IMU

Method	Avg. (m)	Circle	Oval	Eight	Impr. (%)
Raw IMU	0.406 ± 0.161	0.254	0.526	0.439	–
Const. mean	0.534 ± 0.073	0.505	0.563	0.535	-31.5
Kalman 1D	0.594 ± 0.350	0.498	0.484	0.836	-46.2
MA-15	1.084 ± 0.265	1.134	0.819	1.298	-166.8

C. Position Error of Machine Learning Methods

Table IV reports the raw-vs-corrected position RMSE for Random Forest and LSTM under pure dead-reckoning ($N = 4$ per trajectory). Both models improved position accuracy on the circle and, to a lesser extent, on the oval trajectory, but degraded sharply on the figure-eight trajectory.

TABLE IV
POSITION RMSE OF RANDOM FOREST AND LSTM, RAW VS. CORRECTED (N=4)

Method	Traj.	Raw (m)	Corr. (m)	Impr. (%)
Random Forest	Circle	0.409	0.319	$+19.1$
	Oval	0.459	0.436	$+3.9$
	Eight	0.669	1.340	-153.5
LSTM	Circle	0.790	0.734	$+5.1$
	Oval	0.487	0.606	-25.7
	Eight	0.725	1.318	-108.2

Fig. 1 shows the estimated circular trajectory for both models against the Vicon reference, averaged over repetitions, illustrating the closer tracking obtained through gyroscope correction under constant-curvature motion.

On an additional straight-line verification trajectory, evaluated through the empirical error-propagation covariance ($N = 4$), Random Forest achieved the lowest final position error (0.052 m) among all six methods, followed by LSTM (0.076 m), both below the raw IMU (0.116 m) and the three classical methods (Fig. 2). On an exploratory zigzag trajectory ($N = 3$), all methods except the constant-mean correction degraded position accuracy, with Random Forest and LSTM showing the largest drops (-377.5% and -241.3%, respectively), consistent with the figure-eight results.

V. DISCUSSION

Both models learned a valid, statistically significant relationship between the input variables and the gyroscope residual (Section IV-A), with a moderate residual R^2 of 0.23 (Random Forest) and 0.21 (LSTM), indicating that a real but partial fraction of the gyroscope error is explainable from the available inputs. Between the two, Random Forest showed a markedly smaller directional bias asymmetry (0.00029 rad/s vs. 0.00245 rad/s for LSTM), plausibly because its explicit lag and rolling-statistics features encode temporal structure directly, reducing the burden on the model to learn this structure from a moderate-sized training set (16 recordings) at a single fixed velocity.

The most consistent pattern across all six evaluated methods was degradation on trajectories with frequent turning-direction changes (figure-eight, zigzag). Because both learning-based corrections rely on historical information (lags/rolling windows for Random Forest, past sequences for LSTM), neither reacts instantaneously to a curvature reversal; the resulting brief lag in correction accumulates through integration over the course of the trajectory. This is consistent with prior findings on Ackermann-configured vehicles, where navigation schemes lacking an explicit turning/steering model exhibit accelerated position-error growth during frequent direction changes [16], [17]. Neither model incorporated an explicit Ackermann kinematic constraint, relying solely on the gyroscope and magnetometer history, which is consistent with their

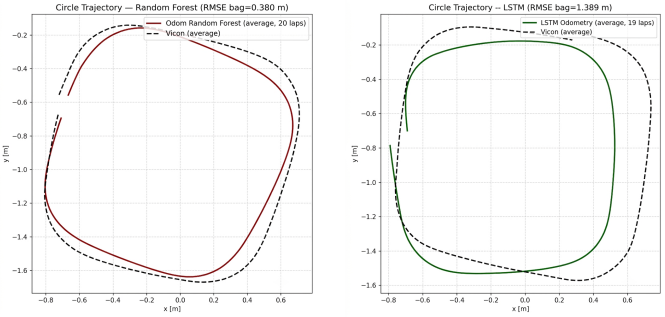


Fig. 1. Estimated circular trajectory vs. Vicon reference (average over repetitions). Left: Random Forest. Right: LSTM.

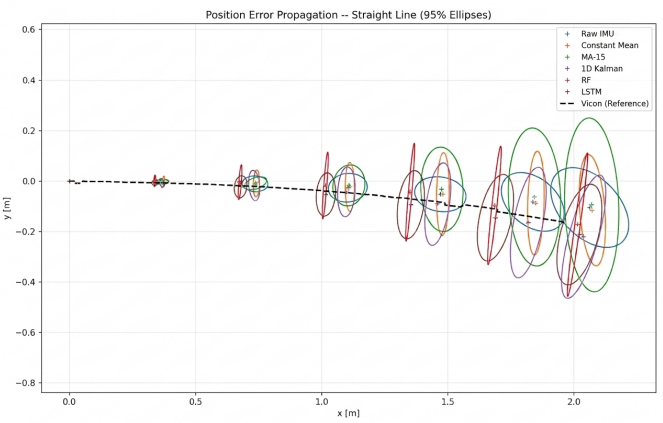


Fig. 2. Position error propagation on the straight-line trajectory, 95% confidence ellipses for the six evaluated methods.

comparatively poor performance precisely in this regime.

The three classical methods degraded position accuracy in nearly every case despite the additional EKF fusion stage available to them, while the AI-based methods improved position accuracy on circle and oval *without* that same fusion support, under a strictly harder, pure dead-reckoning evaluation. This asymmetry strengthens the interpretation that the position-error reduction achieved by Random Forest and LSTM on constant-curvature trajectories is attributable specifically to the quality of the learned gyroscope correction, rather than to compensation by a downstream fusion filter.

From a deployment perspective, Random Forest offers practical advantages over LSTM beyond raw accuracy: inference reduces to traversing pre-trained decision trees, a low-cost operation relative to the 10 Hz gyroscope sampling rate, and it does not require a deep-learning framework on the embedded platform, only lightweight libraries such as scikit-learn, reducing software dependencies and deployment complexity on a device such as the Jetson Orin Nano.

VI. Conclusion

This work shows that it is possible to improve mobile robot odometry by intervening on a single low-cost sensor, the gyroscope, without building or modifying a full sensor-fusion stage. Random Forest and LSTM improved both the instantaneous prediction of gyroscope bias and, on constant-turning-direction trajectories, the integrated position error relative to the uncorrected raw IMU, outperforming all three classical reference methods, which degraded position accuracy in nearly every evaluated case. This improvement was obtained under pure dead-reckoning, without any additional fusion stage to mask correction errors, which strengthens its attribution to gyroscope correction quality alone. However, this benefit did not hold on the figure-eight trajectory, where frequent turning-direction changes caused substantial degradation for both learning-based models, an outcome consistent with the absence of an explicit Ackermann kinematic constraint in either model. Future work includes incorporating Ackermann kinematic constraints as explicit model inputs, extending the training dataset to multiple operating velocities beyond the 0.6 m/s used here, exploring a wider range of model hyperparameters, and increasing trajectory diversity during training to improve generalization to previously unseen turning patterns.

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